• DRIVEABILITY—ALTERNATIVE FUEL VEHICLE COMPRESSED NATURAL GAS TANK INSPECTION • FUEL—ALTERNATIVE FUEL VEHICLE COMPRESSED NATURAL GAS TANK INSPECTION	Article No. 02-5-3

FORD: 1996-2000 CONTOUR
1996-2002 CROWN VICTORIA, E SERIES
1998-2002 F-150

ISSUE

Federal Law requires issuance of guidance for CNG tank inspection. The National Highway Traffic Safety Administration (NHTSA) has mandated that natural gas fuel tanks for vehicles produced after December 2nd, 1996 be inspected for damage or deterioration every 36 months, or 36,000 miles, whichever comes first, or after a fire or accident.

ACTION

It is the customer's responsibility to get their CNG fuel containers inspected by 3 years/36,000 miles and to be in compliance with FMVSS requirements and industry standards.

SERVICE INFORMATION

This is a relatively new requirement that applies only to fuel tanks that are classified as pressure vessels. Although it is new to the automotive industry, the requirement for periodic inspections of pressure vessels is a long established practice in the compressed gas industry.

The purpose of this TSB is to provide an overview of the CNG tank inspection procedures, to clarify the criteria for rating damage to natural gas tanks, and to identify the level of damage requiring tank replacement.

What is involved with the inspection?

The natural gas vehicle industry standard is Basic Requirements for Compressed Natural Gas Fuel Containers - ANSI/IAS NGV2. Paragraph 4.1.4 of this standard refers to the Compressed Gas Association (CGA) pamphlet C-6.4 (Methods for External Visual Inspection of Natural Gas Fuel Containers and Their Installations) (Figure 1). C-6.4 lists the visual inspection requirements in detail including the examination of the:

- Installation
- Containers
- Valves and relief devices
- Labeling

C-6.4 requires that an accurate and reliable written report be provided for each container inspected and that an inspection label be placed on each container stating:

- Date the container was inspected
- Inspection agency

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What is required to conduct a cylinder inspection?

- A copy of Compressed Gas Association (CGA) pamphlet C-6.4 (Methods for External Visual Inspection of Natural Gas Fuel Containers and Their Installations). This publication provides information and procedures for conducting visual inspections of natural gas fuel containers. It also contains industry general container inspection pass/fail criteria limits.
- A copy of Compressed Gas Association (CGA) pamphlet C-6.2 (Guidelines for Visual Inspection & Requalification Fiber Reinforced High Pressure Cylinders) (Figure 2). This publication is not absolutely required, but it is highly recommended if type 2, 3 or 4 cylinders are being inspected. C-6.2 provides more descriptive detail regarding what is acceptable/non-acceptable damage to fiberglass reinforcement of type 2, 3 or 4 cylinders.

C-6.2 and C-6.4 can be purchased through the CGA located at:

Compressed Gas Association
Compressed Gas Association 4221 Walney Road, 5th Floor Chantilly, VA 20151-2923 703-788-2700 Website - www.cganet.com

- A copy of Gas Research Institute (GRI) publication GRI-97/0250 (Natural Gas Vehicle Cylinder Care and Maintenance Handbook) (Figure 3). This publication, also, is not absolutely required, but is highly recommended. GRI publication 97/0250 provides a more detailed overview of cylinder inspection procedures than do the other publications, and is especially useful for training of cylinder inspectors.

GRI-97/0250 can be purchased through the Gas Technology Institute (GTI) located at:

Gas Technology Institute
Gas Technology Institute 1700 South Mount Prospect Road Des Plaines, IL 60018 847-768-0500 Website - www.gastechnology.org

- Cylinder Supplier's Re-Inspection Criteria. Basic Requirements for Compressed Natural Gas Fuel Containers - ANSI/IAS NGV2, paragraph 4.1.4, states that, "requirements for re-qualification by inspection or testing during the service life shall be specified by the container designer." Accordingly, most cylinder manufacturers publish unique and specific re-inspection criteria that is closely tailored to their own cylinder designs. In the event of a conflict between the manufacturer's re-inspection criteria and C-6.4 criteria, the manufacturer's criteria shall take precedence. The C-6.4 criteria is industry general, and is typically more conservative than the manufacturer's criteria. This re-inspection criteria can be obtained from the manufacturer at the address printed on the cylinder label or at addresses listed at the end of this document.
- A High Intensity Light that is capable of illuminating all surfaces of the cylinder(s) to be inspected is required.
- Angled inspection mirrors will be required to view cylinder surfaces that are partially concealed in installation.
- Various hand tools will be required to remove covers, shields, etc. so that the external cylinder surfaces, brackets, valves, etc. can be viewed.
- A torque wrench is required to verify that mounting bracket bolts are properly tightened, and to re-install covers, shields, etc. that are removed to aid in the inspection.
- Depth Gages are required to determine the depth of any cuts, pits, and abrasions that may be encountered.

Who can do the inspections?

C-6.4 states that a qualified inspector should perform the visual inspection of a natural gas vehicle fuel container. C-6.4 states that a qualified inspector must have one of the following:

- Have a minimum of two years experience conducting container inspections
- Be supervised by someone with two years experience
- Be approved by the container manufacturer, or
- Be certified as an inspector by an appropriate organization (e. g., CGA)

C-6.4 also specifies that a qualified inspector must also have:

- Knowledge of the types of containers used in CNG vehicle systems, and damage allowances for each type
- Understanding of inspection requirements, tests, procedures, etc

Therefore, a qualified inspector does not have to go through any particular training or certification process as long as they can meet the criteria listed above and possess the level of knowledge or resources specified in the Compressed Natural Gas Association requirements C-6.4 (summarized above). An inspector could be a dealership technician, a utility technician, or anyone else that qualifies according to C-6.4. However, it is highly recommended that inspectors become certified, as Ford requires that final inspection reports must be signed-off by a certified inspector for warranty or other work paid for by Ford. To locate a certified inspector in your area, visit the CSA International website at:

- www.csa-international.org/certification/gas_inspectors

What is the CSA certification process?

The CSA certification process is a certification process for CNG container inspectors. This is the preferred way to become qualified to inspect CNG containers but is not the only way. To be come certified as a CNG cylinder inspector by CSA, an applicant must complete a 2-day training course and successfully pass the final examination given at the end of the course. Certification classes are taught periodically at several locations. At the present time ,the normal tuition costs vary between $250 and $300 per student. This normally includes the cost of individual training manuals. With prior notice, it is also possible to arrange for on-site certification classes at a location of your choice. Those desiring such on-site classes must pay instructor's travel expenses in addition to individual tuition costs.

To learn about or enroll in up-coming certification classes or to learn how to arrange for a special on-site class, please contact CSA International at one of the following addresses:

CSA International

CSA International
8501 East Pleasant Valley Road
Cleveland, Ohio, USA, 44131-5575
(216) 524-4990
Website -
www.csa-international.org/certification/gas_inspectors/

CSA International

CSA International
5970 Fairview Road Suite 416
Charlotte, North Carolina, USA, 28210
(704) 552-5125
Website -
www.csa-international.org/certification/gas_inspectors/

Cylinder Services Inc.

Cylinder Services Inc.
1427 S 70th St.
Milwaukee, WI. 53214
(414) 479-0500

What does all this mean to the dealer or customer?

It is the customer's responsibility to get their CNG fuel containers inspected by 3 years 36,000 miles and to be incompliance with FMVSS requirements and industry standards.

Cylinder Damage Classification Overview

There are three classifications of compressed fuel tank damage, as identified by the Compressed Gas Association (CGA): Levels One, Two and Three.

- **Level One** damage is minor, and does not require repair.
- **Level Two** damage is more serious. The fuel tank must be removed from the vehicle and repaired according to the tank manufacturer's guidelines or the C-6.4 criteria before it can be certified as safe.
- **Level Three** damage is damage which cannot be repaired. Any fuel tank exhibiting Level Three damage must be removed from service and disposed of, as described later in this TSB.

The complete definitions of the three damage groupings can be found in the CGA C-6.4.

Damage Classification

Level One damage

- **Level One** damage is any cut, scratch, gouge or pit which is less than 0.254 mm (0.010 inch) in depth.

Level Two damage

Level Two damage is Cuts, scratches, gouges or pits which are greater than 0.254 mm (0.010 inch) in depth. Level two damage requires repair. Other instances of Level Two damage include:

- Minor dents in the tank
- Evidence of minor corrosion or chemical attack
- Minor abrasions
- Peeling paint on the tank coating
- Suspected gas leaks in lines or connections
- A label which lists only the manufacturer and the tank's serial number, but does not have a build or service date

Level Three damage

Level Three damage is more severe than Level Two damage, usually cannot be repaired, and normally requires that the cylinder be removed from service. Definition of level three damage varies between manufacturers. Consult the tank manufacturer's service re-inspection criteria for the specific natural gas vehicle you are inspecting. Other types of Level Three damage include:

- Dents greater than 50.8 mm (2 inches) in diameter
- Dents 15.9 mm (0.625 inches) deep
- Any bulging or bowing
- Any stress corrosion cracking
- Any gas leaks
- Any excessive abrasions
- A missing or non-legible label
- Damage due to fire or excessive heat

Special Notes:

1. Hoop-wrapped type 2 cylinders are likely to exhibit a condition known as "Banding," which can be described as radial or circumferential cracks appearing in the fiberglass wrap. These cracks are a natural part of the cylinder manufacturing process and should not be confused with actual fiber breakage or damage. Banding is not normally considered to be serious. Should a question regarding what is acceptable/non-acceptable banding, please refer to the cylinder manufacturer's re-inspection criteria, or consult a certified inspector.
2. When measuring depths of cuts, scratches, scrapes, gouges or pits, it is important to thoroughly clean the surfaces around the area being measured to insure as accurate a measurement as possible. Cuts, scratches, gouges or other damages to the fiberglass wrap will most frequently require only a small amount of cleaning. Corrosion pits in the metal may require more extensive and careful cleaning away of any red rust or other corrosion residue in the area of the pit.
3. When taking measurements, the inspector should be careful to hold the tool or gage as straight and level as possible, especially when measuring curved surface areas. Should a question arise regarding acceptable/non-acceptable methods of measurement, please refer to the cylinder manufacturer's re-inspection criteria, or consult a certified inspector.

Inspection Procedure

- Refer to the appropriate tank inspection procedures in the vehicle's Workshop Manual and the FCSD Compressed Natural Gas Tank Inspection Procedure Video.
- If Level Three damage is confirmed, the facility performing the inspection is required to notify the vehicle owner that the tank is recommended for replacement. Permission to replace the tank must be obtained - in writing and dated - from the vehicle owner. If the customer refuses to have the condemned tank replaced, the refusal must be well-documented in the service report or invoice.
- The vehicle's Workshop Manual includes procedures for removing the tank from service. These include verifying that the tank is empty, removing the tank solenoid valve and mounting strap assemblies from the tank, and purging the tank with compressed air for 30 seconds. Finally, a 12.7 mm (0.50 inch) hole is to be drilled in the sidewall of the tank.

Documenting a Cylinder Inspection

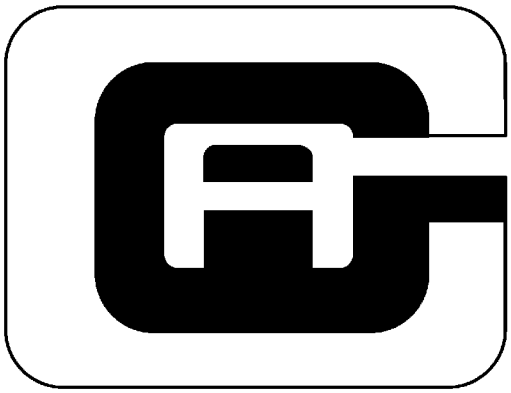
An inspection report needs to be filled out. The CGA has an example of this form in document C-6.4. It will ask for the technician's name, the date, the vehicle, the tank manufacturer, the part and serial number of the tank, and other important information. One copy is to be given to the customer and one is to be included in the vehicle's service records (Figure 4).

OTHER APPLICABLE ARTICLES: NONE
WARRANTY STATUS: INFORMATION ONLY
OASIS CODES: 404000, 490000, 690000, 698298

CGA C-6.4—1998

METHODS FOR EXTERNAL VISUAL INSPECTION OF NATURAL GAS VEHICLE (NVG) FUEL CONTAINERS AND THEIR INSTALLATIONS

COMPRESSED GAS ASSOCIATION, INC.



TB-6563-A

Figure 1 - Article 02-5-3

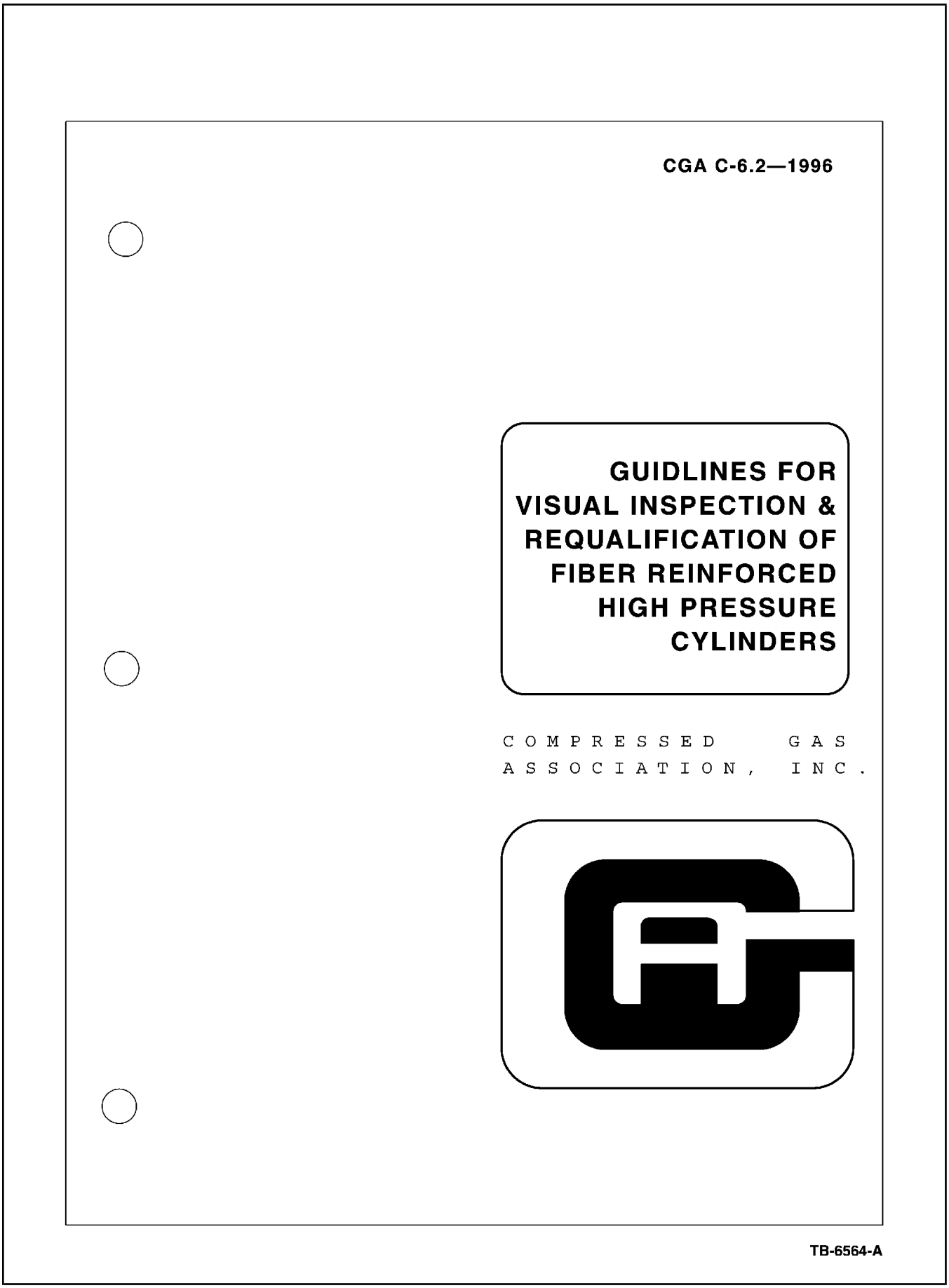


Figure 2 - Article 02-5-3

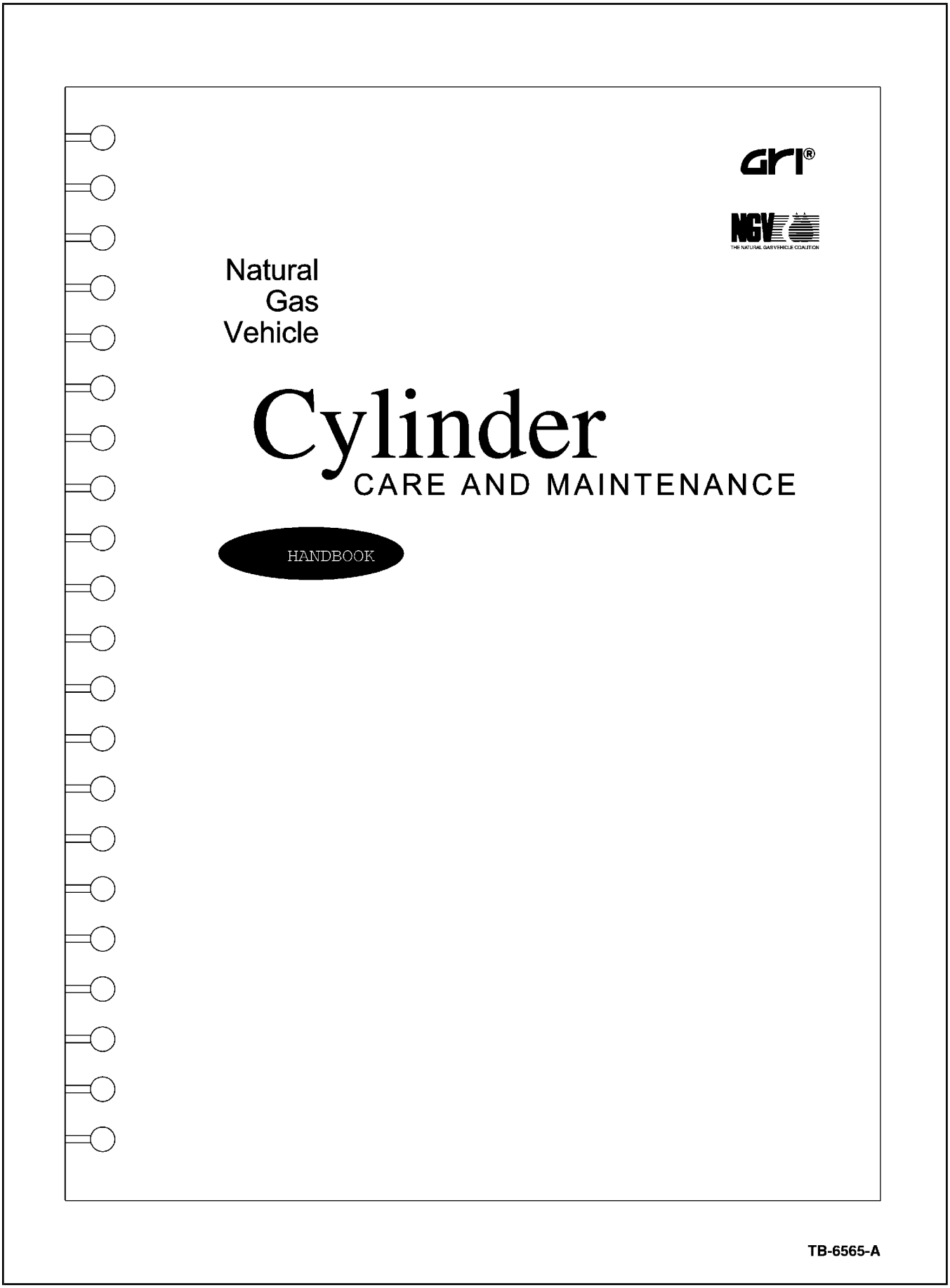


Figure 3 - Article 02-5-3

Appendix A—Sample inspection form

Vehicle Inspector Name: _____
Vehicle Inspector Organization: _____
Date: _____
Vehicle Mileage: _____
Vehicle Make and Model: _____
Vehicle Identification Number (VIN): _____
Manufacturer: _____
Part Number: _____
Serial Number: _____
Location: _____

NOTE: An examination form must be filled out for each fuel container in an installation

Y	N		Examination
		1	Is the container and mounting bracket area clean, free of dirt,and ready for examination?
		2	Is the container free from evidence of fire or exposure to extreme temperatures?
		3	Is the container free from indications that the container has been involved in an accident?
		4	Has the owner been questioned about any conditions or incidents that may have caused damage to the container? (Report adverse findings below).
		5	Is the installation in compliance with applicable regulations? (e.g., NFPA 52 or B149.4 [6, 15])
		6	Is this inspection being conducted prior to the expiration of the container service life?
		7	Are the container service pressure markings greater than or equal to the vehicle service pressure markings?
		8	Is there a minimum of 0.5 inches (12.7 mm) clearance around the container when mounted? (A minimum standoff of 0.375 inches (9.52 mm) is recommended for shields)
		9	If containers are enclosed in the vehicle, are they properly vented external to the vehicle?
		10	Are fuel and vent lines properly and securely attached to the vehicle?
		11	Is the rubber pad between the mounting bracket and fuel container in place and in good condition?
		12	Is the container firmly restrained by the brackets? (i.e., no rocking or looseness)
		13	Are the bolts that secure the brackets to the vehicle present and right?
		14	Is the mounting bracket in good condition, and not bent or deformed?
		15	Is the vehicle free of damage where the mounting brackets are attached?
		16	Are bolts on brackets or straps torqued to proper levels?
		17	Are the valve and/or relief device assemblies free of damage?
		18	Are the valves and relief devices tightly seated? (Caution: Do not loosen valves or relief devices while the tank is pressurized)
		19	Are the interfaces between the valves or relief devices and the container free of leaks?
		20	If there are cuts, gouges, or abrasions present, are they less than 0.01 inch (0.245 mm) deep?
		21	Is the container free of impact damage? (e.g., surface discoloration, cracked resin, chipping, loose fibers)
		22	Is the container free of surface dents?
		23	Is the container free of rust, corrosion, or etching of the outer surface?
		24	Is the container surface free of discoloration?
		25	Is the external paint, composite layer, or metal surface free of bubbles or bulges?
		26	Is the relief device (PRD) in good condition with no visible extrusion of eutectic material?
		27	Are all relief devices in place?
		28	Are brackets or straps free of corrosion?
		29	Has the area under the straps been examined?
		30	Has a new examination sticker or stamp/mark been applied?

Summary of examination and description of any damage of adverse findings:

Rework or replace brackets or other components as follows:

Recommended disposition for fuel container		
	1	Rework fuel container as follows:
	2	Send fuel container to manufacturer for further inspection as follows:
	3	Remove fuel container from service and destroy
	4	Return fuel container to service

Signature of inspector: _____

TB-6566-A

Figure 4 - Article 02-5-3